

Delivery Detour

Mutanguha runs a delivery-robot service on a city grid. A robot starts at $(0, 0)$ and is already programmed with a list of N instructions $S[0], S[1], \dots, S[N-1]$. Each instruction moves the robot one unit: from (x, y) , the instruction L moves it to $(x-1, y)$, R to $(x+1, y)$, D to $(x, y-1)$, and U to $(x, y+1)$.

The robot's firmware allows only one kind of change: cancelling **one contiguous block** of instructions, possibly empty. In other words, pick two integers l and r such that $0 \leq l < r < N$ and replace S with $S[0], S[1], \dots, S[l], S[r], S[r+1], \dots, S[N-1]$. The robot then executes the remaining instructions in their original order. The manufacturer charges a cancellation fee for every cancelled instruction, depending on its direction: C_L for L, C_R for R, C_D for D, and C_U for U.

Just before departure, Mutanguha learns that the customer is actually waiting at (X, Y) .

Help Mutanguha find the minimum total fee so that the robot ends at (X, Y) , or determine that no block of cancellations achieves this.

Implementation Details

You should implement the following procedure:

```
int64 minimum_fee(string S, int32 X, int32 Y, int32 CL, int32 CR,
                  int32 CD, int32 CU)
```

- S : a string of length N over the characters LRDU, the programmed instructions.
- X, Y : the customer's location.
- CL, CR, CD, CU : the cancellation fee of a single L, R, D, U instruction, respectively.
- This procedure is called exactly once.
- This procedure should return the minimum total fee so that the robot ends at (X, Y) , or -1 if this is impossible.

Constraints

- $1 \leq N \leq 200\,000$
- $-N \leq X, Y \leq N$
- $0 \leq C_L, C_R, C_D, C_U \leq 10^9$

Subtasks

Subtask	Score	Additional Constraints
1	15	$N \leq 2\,000$
2	10	$C_L = C_R = C_D = C_U = 0$
3	20	S contains only R and U
4	20	S contains only L and R, and $Y = 0$
5	15	$C_L = C_R = C_D = C_U = 1$
6	20	No additional constraints.

Examples

Example 1

Consider the following call:

```
minimum_fee("RUURLLL", -1, 0, 2, 3, 3, 4)
```

Executing all seven instructions moves the robot to $(-1, 2)$, which misses the customer located at $(-1, 0)$.

If Mutanguha cancels the block UU, the robot executes R, R, L, L, L and reaches the target destination at $(-1, 0)$. The fee for this cancellation is $C_U + C_U = 8$.

While cancelling the larger block UURL also places the robot at $(-1, 0)$, its total fee is higher: $4 + 4 + 3 + 2 = 13$. Because no cancellation sequence yields the correct destination at a cost below 8, `minimum_fee` should return 8.

Example 2

Consider the following call:

```
minimum_fee("RR", 0, 1, 1, 1, 1, 1)
```

No block of cancellations can make the robot end at $(0, 1)$, as the robot never moves vertically. Therefore `minimum_fee` should return -1 .

Example 3

Consider the following call:

```
minimum_fee("LU", -1, 1, 7, 7, 7, 7)
```

The robot already ends at $(-1, 1)$, so Mutanguha should cancel the empty block and pay nothing. Hence `minimum_fee` should return 0.

Sample Grader

Input format:

```
N X Y
CL CR CD CU
S
```

Output format:

```
F
```

Here F denotes the return value of `minimum_fee`.